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# ODS-EA: An objective to decision space-based evolutionary algorithm for high-dimensional feature selection

Mingming Xia [a,b,c], Lei Zhang [d], Kaixuan Li [a,b], Fan Cheng [a,b,c,*]

[a] State Key Laboratory of Opto-Electronic Information Acquisition and Protection Technology, Anhui University, 230039, Hefei, China
[b] the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, 230039, Hefei, China
[c] School of Artificial Intelligence, Anhui University, 230039, Hefei, China
[d] School of Computer Science and Technology, Anhui University, 230039, Hefei, China

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ABSTRACT

Evolutionary algorithms (EAs) have shown their competitiveness in solving feature selection problems. However, due to the "curse of dimensionality", it is quite challenging for EAs to tackle the high-dimensional feature selection (HDFS). To this end, recently, several efforts have been made to design EAs for HDFS. Unlike existing works that solve the challenge from the decision space, in this paper, we tackle the issue from a new perspective: objective space. Specifically, an objective to decision space-based evolutionary algorithm (ODS-EA) is proposed for HDFS, where the feature subsets are obtained by two-stage search in the objective space and the decision space, respectively. In the first stage of ODS-EA, a $(1 \times 1)$ objective space is first constructed for feature selection. Then, a subregion-based evolutionary strategy is suggested to search in the objective space, which can solve the "curse of dimensionality" effectively and select the latent good feature combinations. In the second stage, the search is performed on the decision space, where a feature contribution-based improving strategy is developed to enhance the solutions in the first stage and achieve the final feature subsets with higher quality. Experimental results on 12 high-dimensional datasets demonstrate the effectiveness and efficiency of the proposed ODS-EA, when compared with the state-of-the-art.

## 1. Introduction

Feature selection (FS) is an important data preprocessing technique. It has been widely applied in different domains such as bioinformatics, text mining, and image processing. By selecting a small number of relevant features, FS can reduce the learning cost and improve the performance of a machine learning algorithm (e.g., the classification algorithm) (Dhal & Azad, 2022; Song et al., 2024; Wang et al., 2022). Due to its importance and wide applications, many FS methods by using different optimization techniques have been developed (Alsahaf et al., 2022; Dokeroglu et al., 2022; Khaire & Dhanalakshmi, 2022; Theng & Bhoyar, 2024). Among them, evolutionary computation (EC) has attracted much focus, since it can obtain the global optimal solution and does not require priori knowledge or assumptions on the data and objective functions (Wang et al., 2021).

The existing EC-based FS algorithms can be divided into various categories from different aspects. Based on their interaction with the learning algorithm, they can be mainly grouped into three categories: 1) the filter-based methods; 2) the wrapper-based methods; and 3) the embedded-based methods. Among them, the wrapper-based methods have received widespread attention due to their classifier-driven feedback mechanism, which can achieve high-quality feature subsets. According to the number of objective functions, the existing EC-based FS algorithms can be mainly grouped into two categories: the single-objective based algorithms (Song et al., 2021; Xue et al., 2015; Yu et al., 2025), and the multi-objective based algorithms (Jiao et al., 2023; Liang et al., 2023; Nguyen et al., 2019; Xu et al., 2024b). Compared with the former, the latter can not only adaptively balance multiple objectives in the FS task, but also provide a set of non-dominated solutions. Thus, multi-objective FS algorithms have received increasing attention (Jiao et al., 2023).

Although the EC-based algorithms have shown their competitiveness in tackling the FS problem, most of them are not suitable for the high-dimensional feature selection (HDFS). It is due to the search space of EAs growing exponentially with the number of features, which is known as the "curse of dimensionality" (Yu et al., 2023). To this end, recently,

* Corresponding author.
*E-mail addresses:* xmm@stu.ahu.edu.cn (M. Xia), zl@ahu.edu.cn (L. Zhang), kxli@ahu.edu.cn (K. Li), chengfan@mail.ustc.edu.cn (F. Cheng).

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many efforts have been made to design EAs for HDFS. For example, in (Guan et al., 2021; Wang et al., 2023; Yang et al., 2022b; Zhao et al., 2024), new evolutionary operators were developed to enhance the search efficiency in HDFS. While in (Ahadzadeh et al., 2023; Gong et al., 2023; Song et al., 2022b, 2020; Tran et al., 2018; Yu et al., 2024), the authors tackled the "curse of dimensionality" by directly reducing the original huge search space. Different high-dimensional feature selection tasks showed that these EC-based HDFS algorithms could obtain feature subsets with high quality.

Despite that, by giving a closer look at these HDFS algorithms, we can find that all of them handle the "curse of dimensionality" problem from the perspective of decision space, since the problem arises from the exponential growth of decision space with increasing dimensions. However, an interesting fact is that this challenge does not emerge in the objective space. Just taking bi-objective FS as an example, where the classification error rate (CER $\in$ [0,1]) and the selected feature ratio (SFR $\in$ [0,1]) are two popular objectives (Liang et al., 2023; Wang et al., 2022; Yu et al., 2024). Obviously, no matter how much the dimension grows, the 2$D$ objective space composed of CER and SFR is always in a $(1 \times 1)$ region. Motivated by this fact, in this paper, we solve the HDFS problem from a new perspective: objective space. Specifically, the main contributions of this paper can be summarized as follows.

1) By viewing feature selection as a bi-objective optimization problem, an objective to decision space-based search framework is suggested for HDFS. In the suggested framework, instead of directly searching in the original huge decision space, the search begins from the perspective of objective space, which can tackle the "curse of dimensionality" effectively and find the latent good feature combinations quickly. Then, from the obtained feature combinations, a new search is performed in the decision space, which can further improve the quality of final feature subsets.

2) Based on the suggested framework, an objective to decision space-based evolutionary algorithm (termed ODS-EA) is proposed. In the first stage of ODS-EA, the $(1 \times 1)$ objective space is divided into multiple subregions, and a subregion-based evolutionary strategy is designed to find the latent good feature combinations from the perspective of objective space. Then, in the second stage, a feature contribution-based improving strategy is devised in the decision space, which further enhances the solutions obtained in the first stage and achieves the final feature subsets with higher quality.

3) The performance of the proposed ODS-EA is empirically evaluated by comparing it with five recently suggested baselines for HDFS on 12 high-dimensional datasets, whose number of features ranging from 1024 to 22283. The comparison results verify the effectiveness and efficiency of ODS-EA. Moreover, the ablation experiments also verify the effectiveness of the suggested evolutionary strategies in ODS-EA.

## 2. Related work on EC-based HDFS algorithms

Although EC-based algorithms have shown their competitiveness in feature selection tasks, they often have difficulty tackling high-dimensional scenarios due to the "curse of dimensionality". To this end, recently, several efforts have been made to design different EC-based algorithms for HDFS, which can be mainly grouped into two types (Cheng et al., 2023): 1) enhancing the search efficiency and 2) reducing the search space. Before we review them in detail, it should be admitted that besides EC-based HDFS algorithms, there also exist many non-EC HDFS algorithms, they have also exhibited good performance in solving the HDFS problem. For example, Wang et al. (2020) proposed a sparse subspace learning based discriminative feature selection algorithm ($S^2$DFS), where the trace ratio objective and structured sparse subspace constraint were integrated to obtain sparse feature subsets on datasets with ten of thousands of features. Chen et al. (2025) developed a multi-granularity fuzzy autoencoder-based feature selection algorithm (FAE). In the developed FAE, a feature selection layer and a coarse-grained loss function were incorporated to solve feature selection problems with high-dimensional data. More non-EC HDFS algorithms can be found in Cao et al. (2025), Dong et al. (2024) and Xue et al. (2023).

### 2.1. The HDFS algorithms by enhancing the search efficiency

The first type of EAs solve the "curse of dimensionality" by developing different evolutionary operators, with which the search could be more efficiently performed on the high-dimensional feature space.

A representative work of this type was SM-MOEA (Cheng et al., 2021). In SM-MOEA, a steering matrix that considered both the importance of features and individuals was first suggested. Then, two new evolutionary operators based on the steering matrix were designed, with which the search was concentrated on a few important feature combinations, and the search efficiency was enhanced greatly.

In line with this, Saadatmand and Akbarzadeh-T (2024) proposed a Jaccard similarity-based HDFS approach (JSEMO). It integrated Jaccard similarity into population initialization, reproduction, and elitism steps, which could improve the quality of the population, and enhance the search efficiency. In addition, a double-weighted KNN (KNN2W) classifier was also suggested to solve HDFS problems with imbalanced distributions.

Xu et al. (2024a) proposed a bi-search evolutionary algorithm (BSEA) for HDFS. In BSEA, a forward search task and a backward search task were introduced to improve the search efficiency in the huge space of HDFS. In addition, an adaptive feature analysis mechanism was also designed to find low-frequency features during the evolution, with which better feature subsets could be achieved.

Ma et al. (2021) proposed a two-stage hybrid ant colony optimization algorithm (TSHFS-ACO) for HDFS. In TSHFS-ACO, the determination of the optimal feature subset size and the search for the optimal feature subset were conducted in two separate stages. Specifically, in the first stage, an interval strategy was used to get the rough optimal feature subset size. In the second stage, an advanced ACO algorithm embedded hybrid model was designed to further enhance the search efficiency, and obtain the final feature subset.

Recently, Wang et al. (2024) developed a PSO-based multi-task evolutionary learning algorithm, termed MEL. In MEL, two cooperative FS tasks were first constructed. Then, by incorporating information sharing between these two tasks, MEL achieves enhanced search efficiency and the good-quality's feature subset. Other EAs that belong to this type could be found in Mostafa et al. (2024), Pan et al. (2023), Zhang et al. (2021), Zhu et al. (2023).

### 2.2. The HDFS algorithms by reducing the search space

Different from the first type of algorithms that still search in the original huge feature space, the search space in the second type is directly reduced. Its intuition is from the following fact. Although the total number of features in HDFS is large, many features are not so important. Thus, by directly ignoring some "unimportant" features, the search is only performed in the space with the remaining features.

A representative work of this type was the VLPSO suggested by Tran et al. (2018). In VLPSO, a variable-length PSO representation was suggested for HDFS instead of the original fixed-length one, where different particles had different and smaller lengths. By using the variable-length scheme, VLPSO directly reduced the search space into a much smaller size and focused its search on smaller and more fruitful feature combinations.

Following this, Zhou et al. (2022) proposed a length-adaptive genetic algorithm with Markov blanket (LAGAM) for HDFS. In LAGAM, a relevance-based length-variable individual encoding scheme was first devised. Then an adaptive length change operator and a Markov blanket-based local search were introduced, with which good feature subsets could be achieved by searching in a greatly reduced space.

Song et al. (2022a) proposed a surrogate sample-assisted PSO algorithm called SS-PSO for HDFS. In SS-PSO, a non-repetitive uniform

sampling strategy and a collaborative feature clustering mechanism were designed to divide the sample and feature spaces, which could reduce the search space greatly. On the reduced feature spaces, an ensemble surrogate construction and management strategy was suggested to obtain high-quality feature subsets.

Ahadzadeh et al. (2023) proposed a simple, fast, and efficient HDFS algorithm (SFE). In SFE, two operators (non-selection and selection operators) were developed to reduce the dimensionality of the dataset. To further improve the performance of SFE, a hybrid algorithm embedded in PSO was proposed to search for the optimal feature subset in the reduced dataset.

Yu et al. (2023) developed a bidirectional dynamic grouping-based HDFS algorithm (BDGMOEA). In BDGMOEA, inspired by the idea of feature grouping, the original HDFS problem was transformed into an FS task with a smaller search space. Then, a bidirectional dynamic adjustment mechanism for feature grouping was presented, which ensured the quality of the final feature subset.

Following BDGMOEA, Yu et al. (2023) further devised a multiform optimization approach based on space transformation (MOFS-MST) for HDFS. In MOFS-MST, two space transformation strategies were suggested to construct two smaller search spaces with different feature combinations. Then, a knowledge transfer strategy between the two smaller spaces was implemented to help each other escape the local optima.

Recently, by constructing different low-dimensional FS tasks, several evolutionary multitasking (EMT)-based algorithms have been developed for HDFS. For example, Chen et al. (2022a,b) proposed two EMT-based PSO algorithms for HDFS. In their first work (PSO-EMT), a low-dimensional FS task was created and co-evolved with the original high-dimensional FS task. Then, two mechanisms (variable-range strategy and subset updating mechanism) were developed to further reduce the search space and maintain population diversity, respectively. In their second work (MTPSO), the high-dimensional FS task was converted into several related low-dimensional FS tasks. By performing knowledge transfer between these low-dimensional FS tasks, the optimal feature subset could be effectively found. More recently, Li et al. (2023) constructed multiple low-dimensional FS tasks by using different filtering methods. Then a competitive swarm optimizer was modified to transfer knowledge among these tasks more efficiently, from which a high-quality feature subset could be obtained. More algorithms that solve HDFS problems by reducing search space could be found in Chaudhuri (2024), Cheng et al. (2023), Liu et al. (2022), Yang et al. (2022a).

These two types of EAs have shown their effectiveness and efficiency in solving the HDFS problem. However, by taking a close look at them, we can find that all of them tackle the “curse of dimensionality” from the perspective of decision space. Different from them, in this paper, a new EA is proposed, which solves the issue of HDFS from the perspective of objective space (See Fig. 1 in Section 3). In the next section, we will present it in detail.

## 3. The proposed method

Before introducing the details of the proposed algorithm, first of all, we put the main notations used in this paper in Table 1. We hope this table could help the readers better understand our algorithm.

### 3.1. Basic idea

The basic idea of the proposed method can be described as follows. For the task of feature selection, we first model it as a bi-objective optimization problem, where classification error rate (CER ∈ [0,1]) and selected feature ratio (SFR ∈ [0,1]) are adopted as two objectives. The lower the values of the two objectives, the better the performance of an FS algorithm. Specifically, the two objectives can be formally defined as

$$\begin{cases} \text{Minimize} & obj_1 = CER(I) \\ \text{Minimize} & obj_2 = SFR(I) \end{cases} \quad (1)$$

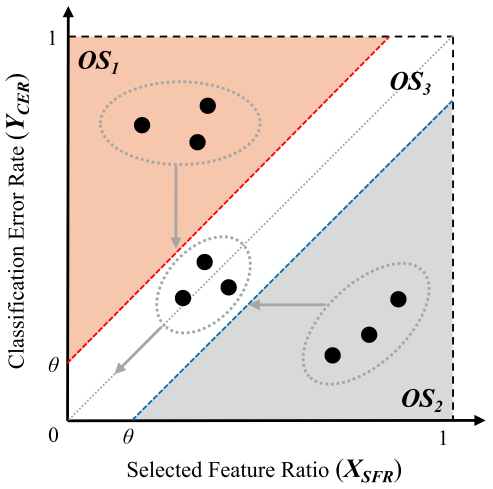



**Fig. 1.** An illustrative example of objective subregion division.

where $I$ denotes a feature subset.

Obviously, in this case, no matter whether the FS task is high-dimensional or low-dimensional, its objective space is always bounded within a $(1 \times 1)$ region. Thus, in the first stage of the proposed method, we focus on this bounded objective space instead of the huge decision space in HDFS. Specifically, we divide the objective space into different subregions, and different evolutionary strategies are designed for each subregion, with which promising feature subsets can be obtained quickly. Then, in the second stage, a feature contribution-based improving strategy is developed to enhance the feature subsets in the first stage, where feature contribution is evaluated from the decision space and guides the evolution towards better directions. In the following, we present in detail the designed strategies in objective and decision spaces, respectively.

### 3.2. Subregion-based evolutionary strategy in the objective space

In the first stage of ODS-EA, a subregion-based evolutionary strategy (SES) is suggested to guide the search toward the promising regions in the objective space. Specifically, the suggested strategy first divides the objective space into multiple subregions, where different subregions have different evolutionary directions. Then, by designing a specific offspring generation strategy for each subregion, the population can gradually move towards better regions in the objective space, which ensures the latent good feature subsets can be obtained effectively.

#### 3.2.1. Objective subregion division

Suppose that the horizontal axis of the objective space is the selected feature ratio ($X_{SFR}$) and the vertical axis is the classification error rate ($Y_{CER}$). The objective space is divided into three subregions as follows, which is shown in Fig. 1. To be specific, we take $Y_{CER} = X_{SFR} + \theta$ (red line in Fig. 1) and $Y_{CER} = X_{SFR} - \theta$ (blue line in Fig. 1) as two dividing lines, where $\theta$ is a parameter to control the size of divided subregions. These two lines divide the whole objective space into three subregions: $OS_1$, $OS_2$, and $OS_3$, each reflecting a different trade-off between objectives. The characteristics of these subregions are summarized as follows.

- $OS_1$ ($Y_{CER} > X_{SFR} + \theta$): In this region (orange region in Fig. 1), the individuals (feature subsets, plotted as black nodes) often have the worse CER values than SFR values. It indicates that these individuals should be guided to move downward, so that individuals with better (i.e., lower) CER values can be searched.
- $OS_2$ ($Y_{CER} < X_{SFR} - \theta$): In this region (grey region in Fig. 1), the individuals often have worse SFR values than CER values. This means

**Table 1**
Main notations used in this paper.

| Symbols | Descriptions |
|---|---|
| $\theta$ | Parameter controlling the subregion size |
| $N_{pop}^1$ | Population size in the objective space |
| $N_{new}$ | Number of newly added individuals in the decision space |
| $Newpop$ | A diverse population of size $N_{new}$ initialized at the start of the search in the decision space |
| $N_{pop}^2$ | Population size in the decision space |
| $n$ | Number of offspring generated per individual in the objective space |
| $Pop^t$ | Population at the $t$-th generation |
| $avgCER^t$ | Average classification error rate of $Pop^t$ |
| $avgSFR^t$ | Average selected feature ratio of $Pop^t$ |
| $rCER$ | Changes in $avgCER$ between two consecutive generations |
| $rSFR$ | Changes in $avgSFR$ between two consecutive generations |
| $thr$ | Threshold for switching from the objective space to the decision space |
| $OS_1, OS_2, OS_3$ | Three subregions in the objective space |
| $gp$ | Guidance position, determined by subregion rules to guide individuals |
| $gs$ | Guidance strength, indicating the influence of $gp$ on individual movement |
| $Pool$ | Candidate feature pool for the offspring generation process in the objective space |
| $rad$ | Radius of the neighborhood around $gp$ |
| $T_{\max}$ | Maximum number of generations |

that these individuals should be guided to move leftward, so that the individuals with better (lower) SFR values can be searched.

- $OS_3$ ($X_{SFR} - \theta \le Y_{CER} \le X_{SFR} + \theta$): In this region (white region in Fig. 1), obviously, the CER and SFR of individuals are relatively balanced. Therefore, the individuals should be guided to move towards the lower left corner, so that the individuals with both better CER and SFR values can be searched.

### 3.2.2. Subregion-based offspring generation

According to the division in the objective space, a subregion-based offspring generation strategy is designed, with which the latent good individuals (feature subsets) can be found effectively. Specifically, for each individual $ind$ in the population $Pop$, we calculate a suitable guiding position $gp$ for $ind$ according to the subregion it belongs to. By moving $ind$ towards $gp$, its offspring are guided towards more promising regions in the objective space. In addition, to prevent over-guidance and maintain exploration, we further introduce the "guidance strength" $gs$ for the guiding position $gp$, which controls the degree of $gp$ on the movement of $ind$ in the objective space.

Let's assume that for the $i$-th individual $ind_i \in Pop$, its CER and SFR in the objective space are denoted as $Y_{CER}^{ind_i}$ and $X_{SFR}^{ind_i}$, respectively. Thus, the following three cases are considered.

***Case 1***: The individual $ind_i$ lies in the $OS_1$ region. According to the guiding rule before, the guidance position $gp$ for $ind_i$ should be set downward to its current position. To this end, $gp$ is set as the intersection point of two lines $X_{SFR} = X_{SFR}^{ind_i}$ and $Y_{CER} = X_{SFR} + \theta$. Therefore, the position of $gp$ in the objective space is denoted as $(X_{SFR}^{ind_i}, X_{SFR}^{ind_i} + \theta)$. In this case, the "guidance strength" $gs$ is determined by the relative distance between $gp$ and $ind_i$ in the $Y_{CER}$ dimension, which is $1 - \frac{Y_{CER}^{gp}}{Y_{CER}^{ind_i}}$.

***Case 2***: The individual $ind_i$ lies in the $OS_2$ region. According to the guiding rule before, the guidance position $gp$ for $ind_i$ should be guided to move leftward to its current position. To this end, $gp$ is set as the intersection point of two lines $Y_{CER} = Y_{CER}^{ind_i}$ and $Y_{CER} = X_{SFR} - \theta$. Thus, the position of $gp$ in the objective space is denoted as $(Y_{CER}^{ind_i} + \theta, Y_{CER}^{ind_i})$. In this case, the "guidance strength" $gs$ is determined by the relative distance between $gp$ and $ind_i$ in the $X_{SFR}$ dimension, and is set to $1 - \frac{X_{SFR}^{gp}}{X_{SFR}^{ind_i}}$.

***Case 3***: The individual $ind_i$ lies in the $OS_3$ region. According to the guiding rule before, the guidance position $gp$ for $ind_i$ should be located lower left to its current position. To this end, $gp$ is set as $(X_{SFR}^{ind_i} \times rnd, Y_{CER}^{ind_i} \times rnd)$, where $rnd \in [0, 1]$ is a random number. In this case, the "guidance strength" $gs$ is determined by the relative distance between $gp$ and $ind_i$ in both the $X_{SFR}$ and the $Y_{CER}$ dimensions, and is set to $1 - \frac{\sqrt{X_{SFR}^{gp\ 2} + Y_{CER}^{gp\ 2}}}{\sqrt{X_{SFR}^{ind_i\ 2} + Y_{CER}^{ind_i\ 2}}}$.

According to the three cases above, for the $i$-th individual $ind_i$, we can generate its offspring by combining it with the guidance position $gp$. Specifically, the generation process of offspring for $ind_i$ is carried out by the following two steps.

*a)* ***Construct the offspring candidate feature pool***. We consider two kinds of features to construct the candidate feature pool $Pool$. One is the features selected from the individual $ind_i$, which are directly put into $Pool$. The other is the features (denoted as $Set_{gp}$) selected from the feature subsets, which are within the niching[1] of the guiding location $gp$. For each feature in $Set_{gp}$, it has a certain probability of being included in $Pool$ according to the "guiding strength" $gs$. Specifically, the larger the value of $gs$, the higher the probability that the feature in $Set_{gp}$ is put into $Pool$. By using the two kinds of features above, we can construct the candidate feature pool $Pool$.

*b)* ***Generate the offspring from the candidate feature pool***. Suppose that for the individual $ind_i$, there are $n$ offspring to be generated from $Pool$. For each offspring $o_i$ ($i = \{1, \cdots, n\}$), the value of each bit in $o_i$ is determined as follows. For the $j$-th bit in $o_i$, if the corresponding $j$-th feature is not in $Pool$, its value is set to 0. Otherwise, a random number $rnd_j \in [0, 1]$ is first generated. Then, if $rnd_j \le X_{SFR}^{gp}$, the $j$-th bit of $o_i$ is set to 1, which indicates the $j$-th feature is included. This operation is repeated for all $D$ bits in $o_i$. After that, an offspring $o_i \in \{0, 1\}^D$ is generated, in which about $X_{SFR}^{gp} \times D$ bits are set to 1. Similarly, this offspring generation strategy is used to generate the remaining $n - 1$ offspring.

After the $n$ offspring are generated for the individual $ind_i$, we need to choose one from them as the individual for the next generation. Since the above offspring generation strategy ensures that all $n$ offspring have similar selected feature ratios, thus, we mainly compare their CER values with that of $ind_i$. Specifically, we choose the offspring ($o_{le}$) with the lowest error rate and replace $ind_i$ by $o_{le}$ if $o_{le}$ has a lower error rate than $ind_i$.

Algorithm 1 gives the pseudocode of the suggested subregion-based evolutionary strategy (SES) in the objective space. Fig. 2 further presents an example to illustrate SES. Suppose that the parameter $\theta$, which controls the subregion size, is set to 0.2. Based on this setting, the objective space is partitioned into three subregions ($OS_1$, $OS_2$, $OS_3$) by two lines $Y_{CER} = X_{SFR} \pm 0.2$ (see Fig. 2(a)). Due to space limitations, we

[1] Here, the radius $rad$ of the niching is set as the median of the distance from all individuals in the population $Pop$ to the $c$-th nearest neighbor.

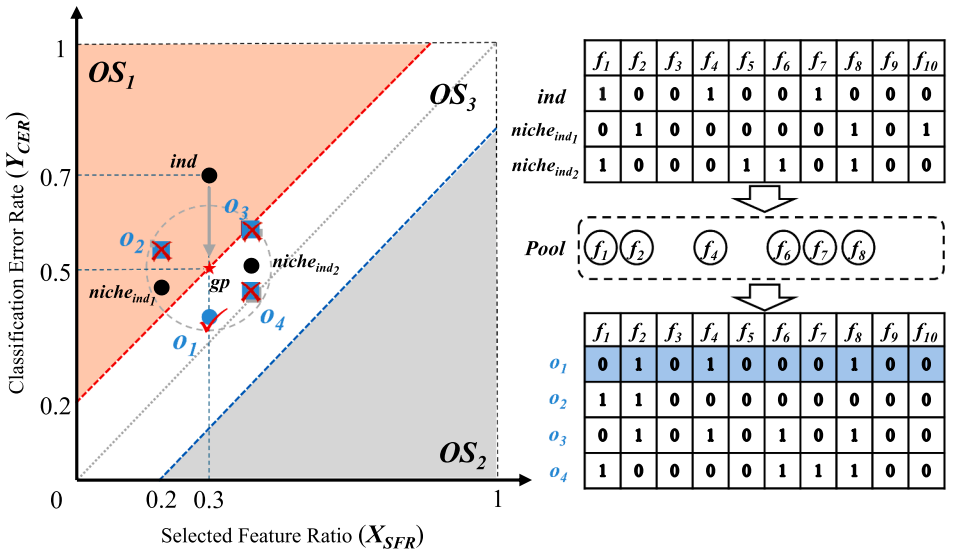



**Fig. 2.** An illustrative example of the subregion-based evolutionary strategy.

**Algorithm 1:** SES.

**Input:** $Pop$: the population; $\theta$: the parameter to control the subregion size; $n$: the number of offspring generated for an individual.

**Output:** $Pop$: updated population for next generation.

1: $\{OS_1, OS_2, OS_3\} \leftarrow$ Divide the objective space by two dividing lines ($Y_{CER} = X_{SFR} \pm \theta$). // See Fig. 1
2: **for all** $ind$ in $Pop$ **do**
3: $[gp, gs] \leftarrow$ Calculate the guidance position and guidance strength for $ind$.
4: $Pool \leftarrow$ Get the offspring candidate feature pool.
5: $o \leftarrow$ Generate $n$ offspring for $ind$ based on $Pool$.
6: $o_{le} \leftarrow$ Select the offspring with the lowest error rate of $o$.
7: $ind \leftarrow$ Replace $ind$ by $o_{le}$ if $o_{le}$ has a lower error rate than $ind$.
8: **end for**
9: **return** $Pop$.

consider a toy dataset with $D = 10$ features. Suppose that an individual $ind$ is located in $OS_1$ and selects the feature set $\{f_1, f_4, f_7\}$. It has a selected feature ratio of $X^{ind}_{SFR} = 0.3$ and a classification error rate of $Y^{ind}_{CER} = 0.7$. According to the rule in Case 1, the guidance position $gp$ (red star) for $ind$ is set as the intersection point of $X_{SFR} = 0.3$ and $Y_{CER} = X_{SFR} + 0.2$, which gives $gp = (X_{SFR}, X_{SFR} + 0.2) = (0.3, 0.5)$. Based on the position $gp$, the guidance strength $gs$ is calculated as $gs = 1 - \frac{Y^{gp}_{CER}}{Y^{ind}_{CER}} = 1 - \frac{0.5}{0.7} \approx 0.29$.

With the $gp$ and $gs$ determined, we can construct a candidate pool $Pool$, which consists of two kinds of features (see the upper part of Fig. 2 (b)). The first kind are the features selected by $ind$, i.e., $\{f_1, f_4, f_7\}$, which are directly included in $Pool$. The second kind of features comes from a candidate set $Set_{gp}$, which consists of features appearing in neighboring individuals within the niche centered at $gp = (0.3, 0.5)$. For each feature $f_j \in Set_{gp}$, a random number $rnd_j \in [0, 1]$ is generated. If $rnd_j \leq gs$, then $f_j$ is added to $Pool$. In this example, suppose that three features $\{f_2, f_6, f_8\}$ are added in $Pool$. Thus, the candidate pool is $Pool = \{f_1, f_2, f_4, f_6, f_7, f_8\}$.

Based on $Pool$, offspring are generated according to step b). We illustrate this process with the generation of $o_1$. For the feature $f_1$, it is in $Pool$. Then, we generate a random number $rnd_1 = 0.56$. Since $X^{gp}_{SFR} = 0.3 < rnd_1$, $f_1$ is not included in $o_1$, and the corresponding bit of $o_1$ is set to 0. For the feature $f_2$, which is also in $Pool$. Its generated random number is $rnd_2 = 0.24$. As $X^{gp}_{SFR} = 0.3 > rnd_2$, $f_2$ is included in $o_1$, and the corresponding bit of $o_1$ is set to 1. While for the feature $f_3$, because it is not in $Pool$, thus $f_3$ is not included in $o_1$, and the corresponding bit of $o_1$ is directly set to 0. Repeating this rule on all $D$ features (bits) in $o_1$, we obtain the offspring $o_1 = (0, 1, 0, 1, 0, 0, 0, 1, 0, 0)$. By applying the same procedure, assume that four offspring are generated near the guidance position $gp$ (see the lower part of Fig. 2(b)), with objective values $(0.3, 0.39)$, $(0.2, 0.52)$, $(0.4, 0.58)$, and $(0.4, 0.45)$, respectively. Among them, $o_1$ achieves the lowest classification error rate ($Y^{o_1}_{CER} = 0.39$). Therefore, $o_1$ is selected to replace $ind$ in the next generation, which is shown in Fig. 2(a). Similarly, for the individuals in $OS_2$ and $OS_3$, offspring are generated by using the same procedure, however, their guidance positions are defined according to the subregion-specific rules (Case 2 and Case 3).

### 3.3. Feature contribution-based improving strategy in the decision space

By evolving with the suggested SES strategy in the objective space, a set of promising feature subsets can be obtained effectively. To further enhance the quality of final feature subsets, a feature contribution-based offspring generation strategy (FCS) is suggested, which is performed from the perspective of decision space and further improves the quality of final feature subsets. Specifically, it first initializes the population by utilizing the individuals obtained from the objective space, and then evolves the population by a newly designed pair of crossover/mutation operators.

#### 3.3.1. Diversity-based population initialization

After the search with the SES, the population has converged to the lower left corner of the objective space quickly. To avoid the possible local optima, $N_{new}$ new individuals are added to the population, which can increase the diversity of the population for subsequent evolution

in the decision space. To this end, the newly added individuals should select features that occur less frequently in the current population.

Specifically, for the $i$-th newly added individual, its $j$-th bit is initialized as follows. First, the ratio $rat_j$ of feature $f_j$ selected in the current $Pop$ is calculated. Then, $rat_j$ is compared with a random number $rnd \in [0, 1]$. If $rat_j < rnd$, it indicates this bit (feature) may occur less frequently in current population. To increase the diversity, the $j$-th bit (feature) of the $i$-th added individual is set to 1; otherwise, it is set to 0. This process is repeated until all $D$ bits of the $i$-th individual are initialized. The same initialization strategy is applied to initialize $N_{new}$ individuals. By adding these new individuals, the diversity of the population is increased. After that, the population $Pop$ has $N^1_{pop} + N_{new}$ (denoted as $N^2_{pop}$) individuals, where $N^1_{pop}$ is the number of individuals when evolving in the objective space.

### 3.3.2. Feature contribution-based offspring generation

After the population $Pop$ evolves in the objective space, a feature contribution-based offspring generation strategy for $Pop$ is designed in the decision space, with which the individuals (feature subsets) with higher quality are obtained. Specifically, a pair of feature contribution-based crossover and mutation operators is developed to generate the offspring.

*a) **Feature contribution-based crossover operator***. In the suggested crossover operator, two parents are first selected from the population $Pop$ by using binary tournament selection. For convenience, the parent with the lower CER value (if the CER value is the same, the one with the lower SFR value is selected) is denoted as $P_{good}$ and the other one is denoted as $P_{bad}$. Then, the suggested crossover operator is performed to generate an offspring $o'$. For the $k$-th feature (bit) $f_k$, the following three cases are considered.

***Case 1***: The feature is selected (unselected) in both parents. It means $f_k$ is considered important (unimportant) by both parents. Therefore, the contribution value of $f_k$ in $o'$ is set to 1 (0).

***Case 2***: The feature is selected in $P_{good}$ but not in $P_{bad}$. It means $f_k$ is more likely to be a feature with high contribution. Thus, the contribution value of $f_k$ in $o'$ is set as follows.

$$Ctr_{f_k} = FI_{f_k} \times (0.5 + 0.5 \times \frac{t}{T_{max}}) \tag{2}$$

***Case 3***: The feature is selected in $P_{bad}$ but not in $P_{good}$. It means the contribution of $f_k$ is not high. Thus, the contribution value of $f_k$ in $o'$ is set as follows.

$$Ctr_{f_k} = FI_{f_k} \times (0.5 - 0.5 \times \frac{t}{T_{max}}) \tag{3}$$

where $FI_{f_k}$ denotes the average CER values of the individuals in $Pop$ that contain $f_k$. $t$ and $T_{\max}$ denote the current generation and the maximum generation, respectively.

Based on the rules above, for each feature (bit), its contribution value in $o'$ is compared with a random number $rnd \in [0, 1]$. If its contribution value is greater than $rnd$, the corresponding bit in $o'$ is set to 1. Otherwise, it is set to 0. This process is repeated until all bits of $o'$ are set.

From the descriptions above, we can find that $FI_{f_k}$ is not evaluated in isolation, but derived from the feature subsets that contain $f_k$. As a result, the complementary effect of $f_k$ with other features is naturally considered. Moreover, since $FI_{f_k}$ is iteratively updated as the population evolves, it dynamically reflects the actual contribution of $f_k$.

*b) **Feature contribution-based mutation operator***. After the crossover operation, we perform the mutation operation on the offspring $o'$. In the suggested mutation operator, the contribution of each feature $f_k$ is evaluated using $FI_{f_k}$. Specifically, two features (bits) are randomly selected from $o'$, and then the following three cases are considered.

***Case 1***: If both features are not selected (i.e., both corresponding bits are 0), the feature with higher contribution is selected, and its corresponding bit is mutated to 1.

***Case 2***: If both features are selected (i.e. both corresponding bits are 1), the feature with lower contribution is removed, and its corresponding bit is mutated to 0.

***Case 3***: If one feature is selected and the other is not (i.e., one bit is 1 and the other is 0), we consider their contributions separately to decide the mutation operation. Specifically, for the bit (feature) with a value of 1, its contribution value is compared with a random number $rnd \in [0, 1]$. If its contribution value is smaller than $rnd$, this bit is mutated to 0 (i.e., the feature is removed). While for the bit (feature) with a value of 0, its contribution value is also compared with another random number $rnd \in [0, 1]$. If its contribution value is greater than $rnd$, this bit is mutated to 1 (i.e., the feature is selected).

Based on the rules above, we can obtain a mutated offspring $o$ from $o'$. By repeating the suggested crossover and mutation operators on all the parents in $Pop$, the same number of offspring are generated for next evolution. The pseudocode of the suggested feature contribution-based improving strategy (FCS) in the decision space is given by Algorithm 2.

Fig. 3 presents an example to illustrate the suggested feature contribution-based crossover and mutation operators. Suppose that two parents are given as: $P_{good}$ = (0, 1, 0, 1, 0) and $P_{bad}$ = (1, 1, 0, 0, 1), which are shown in the middle of Fig. 3(a). We assume the feature importance values $FI$ = (0.85, 0.62, 0.78, 0.55, 0.91), the current generation $t$ = 30, and the maximum generation $T_{\max}$ = 100. Based on the suggested crossover operator, the contribution value of $f_2$ is set to 1, and that of $f_3$ is set to 0, according to Case 1. While feature $f_4$ is only selected by $P_{good}$, thus we obtain its contribution value by Formula (2) in Case 2. Its value is 0.36. Features $f_1$ and $f_5$ are only selected by $P_{bad}$, thus we calculate their contribution values by Formula (3) in Case 3. Their values are 0.3 and 0.32, respectively. Once the contribution value of each feature is obtained, the selection of each feature is determined by comparing its contribution value to a randomly generated value. Just taking $f_1$ as an example, if the random value is 0.75, which is greater than its contribution value (0.30), its corresponding bit in the offspring $o'$ is set to 0. Finally, the offspring after the crossover operation is $o'$ = (0, 1, 0, 0, 1), which is shown in the bottom of Fig. 3(a).

After the crossover operation, the mutation operation is performed on offspring $o'$ by randomly selecting a pair of features, with the three possible cases shown in Fig. 3(b). According to the mutation rules, in Case 1 (assume the randomly selected pair is $\{f_1, f_3\}$), since neither feature in $\{f_1, f_3\}$ is selected in $o'$, then corresponding bit of the feature with higher contribution value ($f_1$) is flipped from 0 to 1, resulting in offspring $o$ = (1, 1, 0, 0, 1). Similarly, in Case 2 (if the randomly selected pair is $\{f_2, f_5\}$), both features in $\{f_2, f_5\}$ are selected in $o'$, then the corresponding bit of the feature with lower contribution ($f_2$) is flipped from 1 to 0, resulting in offspring $o$ = (0, 0, 0, 0, 1). In Case 3 (when the selected pair is $\{f_2, f_4\}$), each feature in $\{f_2, f_4\}$ is independently considered based on its contribution value and a random value. Since the contribution value of $f_2$ (0.62) is less than a random value (0.71), its bit is flipped from 1 to 0. While the contribution value of $f_4$ (0.55) is greater than a random value (0.4), so its bit is flipped from 0 to 1. Therefore, in Case 3, the resulting offspring is $o$ = (0, 0, 0, 1, 1).

## 3.4. The general framework

Based on the two suggested evolutionary strategies in the objective space and the decision space, a multi-objective evolutionary algorithm called ODS-EA is proposed for HDFS. The flowchart of ODS-EA is illustrated in Fig. 4. According to the flowchart, Algorithm 3 further gives the pseudocode of ODS-EA, which consists of three phases: (1) the initialization phase, (2) the main loop phase, and (3) the output phase. Lines 1–3 are the initialization phase. Lines 4–16 are the main loop phase. In each generation of the main loop, the changes in the average CER (classification error rate) values and SFR (selected feature ratio) values between $Pop$ and the previous generation are first calculated (Lines 5–6). If the changes are greater than a predefined threshold $thr$, the evolution of ODS-EA is in the first stage: search in the objective space with the suggested subregion-based evolutionary strategy (Line 11). Otherwise, the evolution of ODS-EA switches to the second stage: search in the decision space with the suggested feature contribution-based improving strategy

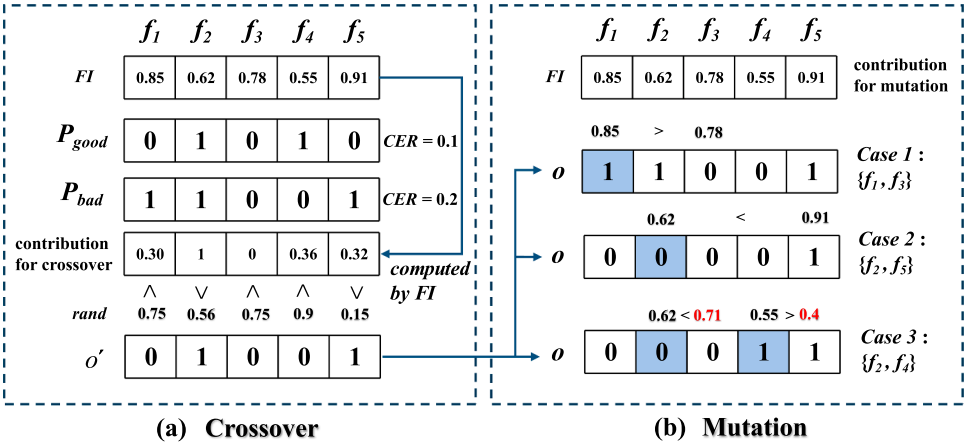



**Fig. 3.** An illustrative example of the suggested feature contribution-based crossover and mutation operators.

**Algorithm 2:** FCS.

**Input:** $Pop$: the population; $N_{new}$: the number of added individuals when evolving in the decision space.
**Output:** $Pop$: updated population for next generation.
1: $Newpop \leftarrow$ Initialize a diverse population of size $N_{new}$. // ***Lines 1–2 occur only when the FCS is performed for the first time***
2: $Pop \leftarrow Pop \cup Newpop$.
3: **for** $i = 1$ to $|Pop|$ **do**
4: $[p_1, p_2] \leftarrow$ Select two individuals from $Pop$ using binary tournament selection.
5: $o' \leftarrow$ Apply the feature contribution-based crossover operator to $p_1$ and $p_2$.
6: $o \leftarrow$ Apply the feature contribution-based mutation operator to $o'$.
7: $O \leftarrow O \cup o$.
8: **end for**
9: $Pop \leftarrow$ Update $Pop$ by $Pop \cup O$.
10: **return** $Pop$.

**Algorithm 3:** ODS-EA.

**Input:** $Data$: the original data; $T_{\max}$: maximum generations; $N^1_{pop}$: population size in the objective space; $N_{new}$: number of newly added individuals in the decision space; $\theta$: the parameter to control the subregion size; $thr$: the threshold for changes in classification error rate and selected feature ratio; $n$: the number of offspring generated per individual in the objective space.
**Output:** the first Pareto front of $Pop^{T_{\max}}$.
1: $t = 1$; $avgCER^0 = 0$; $avgSFR^0 = 0$.
2: $InStage1 = true$.
3: $Pop^1 \leftarrow$ Randomly initialize a population with the size of $N^1_{pop}$.
4: **while** $t \leq T_{\max}$ **do**
5: $[avgCER^t, avgSFR^t] \leftarrow$ Calculate the average error rate and selected feature ratio of $Pop^t$.
6: $[r_{CER}, r_{SFR}] \leftarrow$ Calculate the changes in $avgCER$ and $avgSFR$ between two consecutive generations.
7: **if** ($r_{CER} < thr$ and $r_{SFR} < thr$) **then**
8: $InStage1 = false$.
9: **end if**
10: **if** $InStage1 == true$ **then**
11: $Pop^{t+1} =$ **SES** ($Pop^t$, $\theta$, $n$). // Algorithm 1
12: **else**
13: $Pop^{t+1} =$ **FCS** ($Pop^t$, $N_{new}$). // Algorithm 2
14: **end if**
15: $t = t + 1$.
16: **end while**
17: **return** the first Pareto front of $Pop^{T_{\max}}$.

(Line 13). Finally, Line 17 is the output phase, the first Pareto front of $Pop$ is returned as the final feature subsets.

### 3.5. Computational complexity analysis

According to the framework of ODS-EA (Algorithm 3), the first part (Lines 1–3) is the initialization phase. Lines 1–2 are the settings of parameters, which have the computational complexity of $O(1)$. Line 3 involves random generation of the initial population with size $N^1_{pop}$. Since each individual is encoded as a binary vector of length $D$, the initialization cost is $O(N^1_{pop} \times D)$. Therefore, the computational complexity of the initialization phase is $O(N^1_{pop} \times D)$.

The second part (Lines 4–16) is the main loop phase. Lines 5–6 calculate the average classification error rate (CER) and selected feature ratio (SFR) for the current population. Their costs are $O(N^1_{pop} \times m^2 \times D + N^1_{pop} \times D)$. Here, $m$ is the number of instances in the dataset. Lines 7–9 are the stage-switching condition check, whose computational cost is $O(1)$. If the algorithm is in Stage 1, the SES procedure (Line 11) is invoked. According to Algorithm 1, the SES consists of objective space division, guidance calculation, and population update. Their costs are $O(1)$, $O(N^1_{pop} \times D)$, and $O(N^1_{pop} \times m^2 \times D + N^1_{pop} \times n)$, respectively. Here, $n$ is the number of offsprings generated per individual. Thus, the tolal complexity of SES is $O(N^1_{pop} \times m^2 \times D + N^1_{pop} \times n)$. If the algorithm switches to Stage 2, the FCS procedure (Line 13) is executed. According to Algorithm 2, FCS consists of newly added individuals initialization, genetic operations, and population update. Their costs are $O(N_{new} \times D)$, $O(N^2_{pop} \times D)$, and $O(N^2_{pop} \times m^2 \times D + (N^2_{pop})^2)$, respectively. Thus, the computational complexity of the FCS is $O(N_{new} \times D + N^2_{pop} \times D + N^2_{pop} \times m^2 \times D + (N^2_{pop})^2)$, which can be simplified as $O(N^2_{pop} \times m^2 \times D + (N^2_{pop})^2)$. Here, $N_{new}$ and $N^2_{pop}$ denote the number of newly added individuals and total individuals of population in the deision space (Stage 2), respectively. Since the main loop repeats for at most $T_{\max}$ times, thus, the total computational complexity of the main loop is $O(T_{\max}(N^1_{pop} \times m^2 \times D + N^1_{pop} \times n + N^2_{pop} \times m^2 \times D + (N^2_{pop})^2)$. Since $N^2_{pop} = N^1_{pop} + N_{new}$ and also $N^2_{pop} = N^1_{pop} \times n$, it can be simplified as $O(T_{\max}(N^2_{pop} \times m^2 \times D + (N^2_{pop})^2))$.

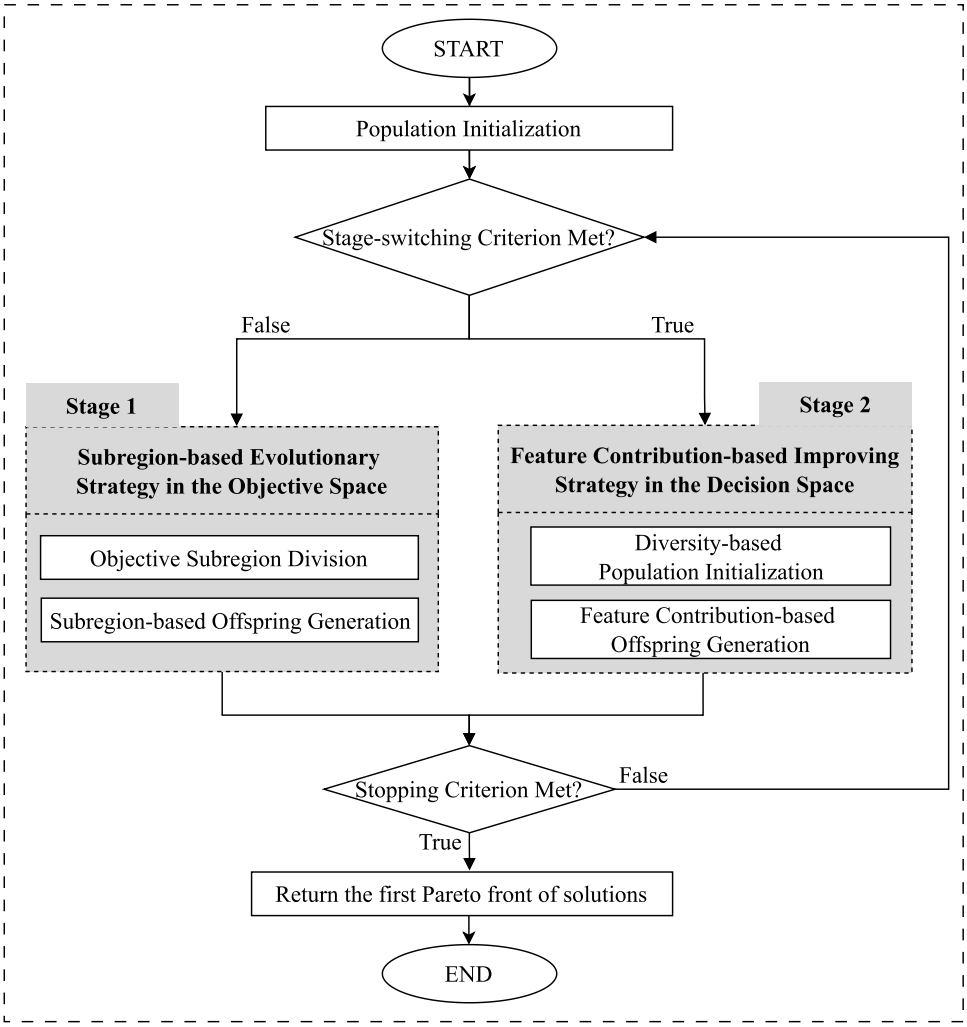



**Fig. 4.** The flowchart of the proposed ODS-EA.

The third part (Line 17) is the output phase. Its complexity is $O((N_{pop}^2)^2)$. Thus, combining the above three parts, the overall time complexity of ODS-EA can be expressed as $O(T_{\max}(N_{pop}^2 \times m^2 \times D + (N_{pop}^2)^2))$.

## 4. Experiments results

### 4.1. Experimental settings

#### 4.1.1. Comparison algorithms

To verify the effectiveness of the proposed ODS-EA, five state-of-the-art EC-based high-dimensional feature selection (HDFS) algorithms, LAGAM (Zhou et al., 2022), SS-PSO (Song et al., 2022a), MEL (Wang et al., 2024), BDGMOEA (Yu et al., 2023), and MOFS-MST (Yu et al., 2024) are adopted as the comparison algorithms. Among them, the first three are single-objective evolutionary algorithms, which are well designed for the HDFS problem. The last two algorithms are recently suggested multi-objective evolutionary algorithms (MOEAs), which have shown good performance in solving the HDFS problem. The computational complexities of ODS-EA and five comparison algorithms are summarized in Table 2.

#### 4.1.2. Datasets

To evaluate the performance of different algorithms, 12 datasets with high dimensions are adopted as the experimental sets, whose dimensions ranging from 1024 to 22283. The detailed statistics are shown in Table 3. For each dataset *Data*, we randomly partition it with 70 % for

**Table 2**
Computational complexities of ODS-EA and five comparison algorithms.

| Algorithms | Computational Complexities |
|---|---|
| LAGAM (Zhou et al., 2022) | $O(T_{\max}(Nm^2n) + mD^2)$ |
| SS-PSO (Song et al., 2022a) | $O(cL + \frac{nD^2}{L} + N\text{Cost}_R + T(NQ + N\text{Cost}_L) + T_sL\text{Cost}_R)$ |
| MEL (Wang et al., 2024) | $O(T_{\max}(Nm^2D))$ |
| BDGMOEA (Yu et al., 2023) | $O(T_{\max}(Nm^2D + gNum \times N))$ |
| MOFS-MST (Yu et al., 2024) | $O(T_{\max}(Nm^2D + N^2))$ |
| ODS-EA [ours] | $O(T_{\max}(N_{pop}^2m^2D + (N_{pop}^2)^2))$ |

**Table 3**
Statistics of the 12 high-dimensional datasets.

| Datasets | #Features | #Instances | #Classes |
|---|---|---|---|
| ORL | 1024 | 400 | 4 |
| colon | 2000 | 62 | 2 |
| warpPIE10P | 2420 | 210 | 10 |
| lymphoma | 4026 | 96 | 9 |
| GLIOMA | 4434 | 50 | 4 |
| DLBCL | 5469 | 77 | 2 |
| TOX_171 | 5748 | 171 | 4 |
| Brain1 | 5920 | 90 | 5 |
| ALLAML | 7129 | 72 | 2 |
| Brain2 | 10,367 | 50 | 4 |
| CLL_SUB_111 | 11,340 | 111 | 3 |
| GLI_85 | 22,283 | 85 | 2 |

**Table 4**
Parameter settings of different algorithms.

| Algorithms | Parameter values |
|---|---|
| LAGAM (Zhou et al., 2022) | $p = \lvert F\rvert/20$, $g = 100$, $p_c = 0.8$, $p_m = 0.2$, $p_l = 0.3$, $\alpha = 0.1$, $\beta = 0.4$ |
| SS-PSO (Song et al., 2022a) | $P_{\max} = 0.5$, $P_{\min} = 0.1$, $T_{\max} = 100$, $N = 100$ |
| MEL (Wang et al., 2024) | $\theta = 0.6$ |
| BDGMOEA (Yu et al., 2023) | $gNum = 10 \times log_2 N_{total\,fea}$, $T_r = 10$, $T_s = T_r \times \lfloor \log_2(N_{total\,fea}/gNum) + 1 \rfloor$ |
| MOFS-MST (Yu et al., 2024) | $\alpha = 1$, $\beta = 3$ |
| ODS-EA [ours] | $\theta = 0.2$, $N^1_{pop} = 25$, $N_{new} = 75$, $thr = 1e\text{-}3$, $n = 4$, $T_{\max} = 100$ |

**Table 5**
The average classification error rates obtained by different HDFS algorithms.

| Datasets | LAGAM | SS-PSO | MEL | BDGMOEA | MOFS-MST | ODS-EA |
|---|---|---|---|---|---|---|
| ORL | 0.193±0.034(+) | 0.224±0.038(+) | 0.205±0.037(+) | 0.179±0.030(+) | 0.195±0.033(+) | **0.132**±0.023 |
| colon | 0.159±0.060(≈) | 0.243±0.101(+) | 0.332±0.129(+) | 0.181±0.080(+) | 0.164±0.065(≈) | **0.142**±0.067 |
| warpPIE10P | 0.175±0.063(+) | 0.167±0.050(+) | 0.198±0.054(+) | 0.059±0.041(+) | 0.052±0.049(≈) | **0.038**±0.028 |
| lymphoma | 0.234±0.094(+) | 0.299±0.107(+) | 0.274±0.103(+) | 0.263±0.100(+) | 0.227±0.097(+) | **0.202**±0.076 |
| GLIOMA | 0.287±0.095(+) | 0.285±0.101(+) | 0.338±0.143(+) | 0.240±0.092(+) | 0.221±0.094(+) | **0.153**±0.086 |
| DLBCL | 0.088±0.093(+) | 0.194±0.094(+) | 0.210±0.111(+) | 0.072±0.089(+) | 0.056±0.072(+) | **0.035**±0.065 |
| TOX_171 | 0.247±0.064(+) | 0.279±0.058(+) | 0.284±0.055(+) | 0.226±0.049(+) | 0.245±0.052(+) | **0.124**±0.039 |
| Brain1 | 0.489±0.111(+) | 0.554±0.139(+) | 0.520±0.127(+) | 0.446±0.154(+) | 0.469±0.125(+) | **0.363**±0.121 |
| ALLAML | 0.093±0.109(+) | 0.131±0.092(+) | 0.168±0.101(+) | 0.045±0.071(+) | 0.038±0.059(+) | **0.011**±0.033 |
| Brain2 | 0.354±0.109(+) | 0.354±0.105(+) | 0.351±0.089(+) | 0.286±0.115(+) | 0.277±0.115(+) | **0.189**±0.095 |
| CLL_SUB_111 | 0.293±0.068(+) | 0.356±0.106(+) | 0.338±0.109(+) | 0.280±0.080(+) | 0.303±0.088(+) | **0.200**±0.050 |
| GLI_85 | 0.160±0.073(+) | 0.254±0.115(+) | 0.223±0.084(+) | 0.153±0.097(+) | 0.153±0.094(+) | **0.095**±0.054 |
| +/ ≈ /− | 11/1/0 | 12/0/0 | 12/0/0 | 12/0/0 | 10/2/0 | − |

training set and the remaining 30 % for testing. During the evolution, the KNN (K = 1) is adopted as the classifier and the leave-one-out cross validation (LOOCV) is used to evaluate the classification error rate of the solution. To reduce variance, 30 independent runs are executed. We report the average results of each algorithm with 30 independent runs. In addition, we present the statistical analysis using the Wilcoxon rank-sum test with a significance level of 0.05. Note that symbols "+", "≈", and "-" denote that the result obtained by the proposed ODS-EA is significantly better, similar, and worse to that of the baseline.

#### 4.1.3. Parameter settings

For the five comparison algorithms, we adopt the recommended parameter values suggested in their original papers. The details about the parameter settings for each comparison algorithm are listed in Table 4. For the proposed ODS-EA, the parameter for controlling the size of divided subregions is set to $\theta = 0.2$. During the evolution in the objective space, the size of population is set to $N^1_{pop} = 25$. While evolving in the decision space, the number of newly added individuals is set to $N_{new} = 75$. When determining whether to switch the evolution from the objective space to the decision space, the threshold for changes in classification error rate and selected feature ratio is set to $thr = 1e\text{-}3$. Additionally, the number of offspring generated by an individual when evolving in the objective space is set to $n = 4$. The maximum number of generations is set to $T_{\max} = 100$.

In addition, for a fair comparison, all the six EAs (including ours) use the same number of evaluations as the stopping criterion, which is set to 10000. All the experiments are conducted on a PC with an Intel Core i7-10700 CPU @2.90GHZ and a total memory of 16 GB.

### 4.2. Experimental results and analysis

The experimental results are first reported from the perspective of feature selection (i.e., classification error rate and feature subset size). In this paper, we employ balanced accuracy to calculate the classification error rate, which is widely used to effectively handle both balanced and imbalanced datasets (Tran et al., 2018; Zhou et al., 2022). Thus, CER is defined as follows.

$$CER = 1 - \frac{1}{c}\sum_{i=1}^{c} TPR_i \tag{4}$$

where $c$ represents the number of classes in the dataset, and $TPR_i$ denotes the true positive rate for class $i$.

In addition, since the proposed ODS-EA is a multi-objective algorithm, the experimental results are also reported from the perspective of multi-objective optimization. Lastly, we verify the effectiveness of the suggested strategies in our ODS-EA.

#### 4.2.1. Comparison from the perspective of feature selection

When measured from the perspective of feature selection, we report the classification error rate, the feature subset size, and the running time of different HDFS algorithms, which are shown in Tables 5–7, respectively. It should be noted that three single-objective algorithms (i.e., LAGAM, SS-PSO, and MEL) naturally obtain a single best solution, which is directly used as the final output. In contrast, multi-objective algorithms (i.e., BDGMOEA, MOFS-MST, and our ODS-EA) produce a set of Pareto-optimal solutions. To this end, we select the solution with the lowest classification error rate from Pareto-optimal solutions as the final output for multi-objective algorithms.

From Table 5, we can find that the proposed ODS-EA achieves the lowest classification error rate on all the datasets. The Wilcoxon statistics also confirm the superiority of ODS-EA, where the proposed method always wins or is comparable to the baselines and never loses once. These results indicate that compared with the state-of-the-art, our ODS-EA can achieve feature subsets with lower error rate.

Table 6 lists the feature subset size obtained by ODS-EA and five comparison algorithms. It can be seen from the table that on most datasets, ODS-EA selects much less number of features than other algorithms. Especially, as the dimension increases, this superiority becomes more distinct. Just taking the "GLI_85" dataset as an example, which has 22,283 features. On this dataset, the proposed method only selects about 46 features on average, while most baselines choose more than hundreds of features.

Table 7 presents the running time obtained by ODS-EA and other comparison algorithms. As shown in Table 7, our ODS-EA has much

**Table 6**
The average feature subset sizes obtained by different HDFS algorithms.

| Datasets | LAGAM | SS-PSO | MEL | BDGMOEA | MOFS-MST | ODS-EA |
|---|---|---|---|---|---|---|
| ORL | 305.1±89.7(+) | 126.6±72.5(+) | 369.2±24.1(+) | **52.6**±11.2(≈) | 136.7±65.4(+) | 56.8±10.7 |
| colon | 36.6±13.0(+) | 101.8±105.8(+) | **6.9**±4.7(−) | 25.0±12.2(+) | 18.1±12.4(+) | 10.9±4.9 |
| warpPIE10P | 414.6±325.0(+) | 123.4±98.2(+) | 823.4±54.2(+) | 35.5±25.0(+) | 36.5±25.2(+) | **16.2**±5.4 |
| lymphoma | 745.7±193.1(+) | 509.7±451.5(+) | 1386.8±99.7(+) | **43.3**±23.1(−) | 81.4±36.6(≈) | 80.9±76.3 |
| GLIOMA | 96.6±104.1(+) | 222.2±213.2(+) | 545.7±748.3(+) | 59.4±24.5(+) | 76.4±37.1(+) | **17.1**±15.1 |
| DLBCL | 49.8±14.8(+) | 47.8±20.7(+) | **16.1**±30.7(≈) | 31.0±29.5(+) | 39.8±26.1(+) | 17.0±18.2 |
| TOX_171 | 315.3±128.6(+) | 193.7±66.7(+) | 2250.8±102.3(+) | 237.5±98.7(+) | 187.9±96.2(+) | **69.0**±33.6 |
| Brain1 | 120.1±68.9(+) | 29.0±43.5(+) | 508.4±888.3(+) | 103.0±41.9(+) | 88.3±71.9(+) | **15.7**±5.3 |
| ALLAML | 94.9±31.1(+) | 21.5±15.3(≈) | 44.3±50.7(+) | 28.6±42.7(≈) | 49.1±32.0(+) | **18.0**±19.7 |
| Brain2 | 92.2±63.3(+) | 97.7±113.5(+) | 269.2±867.9(+) | 129.6±67.1(+) | 193.9±65.0(+) | **18.3**±8.4 |
| CLL_SUB_111 | 300.1±276.6(+) | 260.4±82.8(+) | 365.3±1143.8(+) | 282.5±149.7(+) | 418.2±199.1(+) | **29.7**±8.3 |
| GLI_85 | 239.1±66.4(+) | 89.4±48.4(+) | 896.2±2366.5(+) | 197.3±162.9(+) | 535.1±263.3(+) | **46.4**±26.8 |
| +/ ≈ /− | 12/0/0 | 11/1/0 | 10/1/1 | 9/2/1 | 11/1/0 | – |

**Table 7**
The average running time (seconds) obtained by different HDFS algorithms.

| Datasets | LAGAM | SS-PSO | MEL | BDGMOEA | MOFS-MST | ODS-EA |
|---|---|---|---|---|---|---|
| ORL | 1323.4±85.5(+) | **18.6**±9.0(−) | 54.8±2.0(+) | 35.2±3.0(+) | 51.2±5.1(+) | 24.0±7.4 |
| colon | 54.6±7.5(+) | 8.2±7.4(+) | 12.1±1.3(+) | 7.5±1.8(+) | 8.7±2.9(+) | **4.9**±1.3 |
| warpPIE10P | 785.5±133.9(+) | 20.7±6.5(+) | 46.0±1.4(+) | 22.3±2.1(+) | 33.2±2.7(+) | **10.9**±3.3 |
| lymphoma | 451.3±67.5(+) | 171.5±23.0(+) | 34.8±1.4(+) | 15.9±1.5(+) | 27.1±3.0(+) | **10.1**±2.4 |
| GLIOMA | 80.1±31.8(+) | 18.3±20.1(+) | 33.0±2.8(+) | 16.3±1.6(+) | 20.1±2.7(+) | **7.6**±1.6 |
| DLBCL | 88.0±14.1(+) | **5.0**±2.5(−) | 31.9±2.5(+) | 24.4±1.6(+) | 24.9±2.5(+) | 12.0±1.9 |
| TOX_171 | 691.9±168.4(+) | **11.7**±5.5(−) | 82.5±3.4(+) | 50.7±3.2(+) | 59.4±5.1(+) | 26.2±8.4 |
| Brain1 | 150.0±49.6(+) | **12.1**±6.5(−) | 51.4±8.1(+) | 29.7±1.6(+) | 36.1±2.6(+) | 14.8±2.4 |
| ALLAML | 89.0±17.6(+) | **3.5**±2.7(−) | 40.5±2.9(+) | 27.4±1.7(+) | 28.7±2.6(+) | 13.7±2.7 |
| Brain2 | 88.1±18.6(+) | **8.1**±3.6(−) | 72.2±4.5(+) | 37.7±1.4(+) | 40.6±3.7(+) | 19.4±1.8 |
| CLL_SUB_111 | 306.1±164.1(+) | **15.9**±4.7(−) | 69.6±12.5(+) | 63.3±2.5(+) | 77.8±4.3(+) | 23.1±6.1 |
| GLI_85 | 313.8±70.5(+) | **13.2**±2.7(−) | 139.2±19.0(+) | 102.3±3.4(+) | 121.6±10.3(+) | 41.1±6.9 |
| +/ ≈ /− | 12/0/0 | 4/0/8 | 12/0/0 | 12/0/0 | 12/0/0 | – |

shorter running time than LAGAM, MEL, BDGMOEA, and MOFS-MST, which indicates higher computational efficiency of the proposed method. In addition, Table 7 also reveals that the running time of SS-PSO is comparable or shorter than ODS-EA. The reason is mainly attributed to the following fact. In SS-PSO, the entire sample set is replaced by a small number of surrogate units. It can reduce the computational cost of particle evaluation in PSO significantly. However, this strategy may lead to inaccurate particle evaluation, which decreases the accuracy of the final obtained particles (feature subsets). As shown in Table 5, the classification error rate of SS-PSO is not only inferior to that of our ODS-EA but also worse than those of LAGAM, BDGMOEA, and MOFS-MST. Moreover, for datasets with moderate dimensionality (such as "lymphoma"), the computational advantage offered by the surrogate mechanism is weakened. As a result, the overall advantage in running time is reduced. In contrast, in the proposed ODS-EA, there are two important and complementary strategies: the subregion-based evolutionary strategy (SES) and feature contribution-based improving strategy (FCS). Specifically, the guidance rules of SES can reduce the selected feature ratio (SFR) more quickly, which tends to be more evident on datasets with moderate or not very high dimensionality. This is because the total number of features is not large, so removing the same number of features could cause a larger drop in SFR. Thus, the feature subsets with smaller size can be obtained, thereby reducing the running time of ODS-EA. Meanwhile, FCS improves the quality of feature subsets by selecting features with higher contributions and removing redundant ones, which leads to lower classification error rates of feature subsets. To summarize, the complementary effect of SES and FCS makes ODS-EA to effectively balance efficiency and effectiveness on the high-dimensionality datasets, such as "lymphoma".

Based on the basis of these analyses from three tables above, we can conclude that as a feature selection algorithm, ODS-EA can achieve the feature subsets with both lower error rates and smaller sizes in a shorter running time. Therefore, it is more suitable for solving high-dimensional feature selection problems.

#### 4.2.2. Comparison from the view of multi-objective optimization

Since the proposed ODS-EA models the HDFS as a multi-objective optimization problem, in this section we evaluate all the MOEAs (BDGMOEA, MOFS-MST, and our ODS-EA) from the view of multi-objective optimization.

Fig. 5 plots the median Pareto fronts of three MOEAs on 12 datasets. It can be observed that compared to other multi-objective HDFS algorithms, the proposed ODS-EA achieves better Pareto fronts on most datasets. Especially, on the datasets with higher dimensions, this superiority becomes more distinct. All these results are consistent with the statistics in Tables 5 and 6, which once again demonstrate the superiority of ODS-EA. In addition, by taking a closer look at Fig. 5, we can find that ODS-EA exhibits more preference for exploring feature subsets with lower error rate. In fact, this preferred search behavior is beneficial, since when evaluating the quality of a feature subset, error rate is more important than the feature subset size (Nguyen et al., 2019; Yu et al., 2023).

Furthermore, we report the performance of different MOEAs in terms of the HV (Zitzler & Thiele, 1998) values, which is a popular metric to measure the convergence and diversity of an MOEA. The comparison results of these MOEAs are reported in Table 8. It can be observed from the table that ODS-EA achieves the highest HV on 11/12 datasets, and is the second highest on the remaining dataset. The win/tie/loss statistics in this table also demonstrate the superiority of our ODS-EA. These results show that as an MOEA based feature selection algorithm, ODS-EA has both good convergence and diversity.

### 4.3. The effectiveness of the suggested strategies in ODS-EA

The above better performance of ODS-EA benefits from its two suggested strategies, which are the subregion-based evolutionary strategy in the objective space and the feature contribution-based improving strategy in the decision space. To verify the effectiveness of these two strategies, we compare the ODS-EA with its two variants:

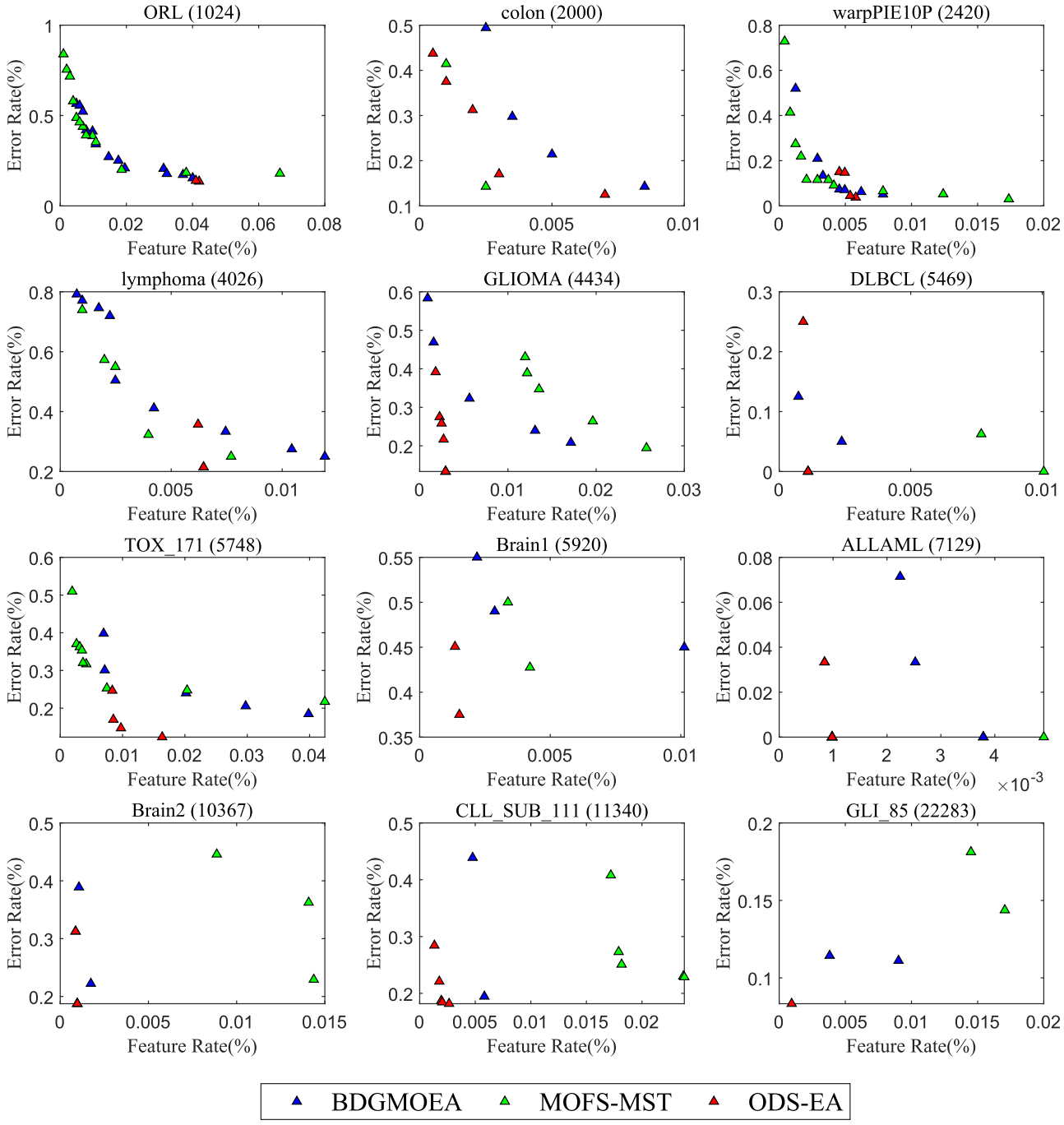



**Fig. 5.** The median Pareto fronts obtained by different MOEAs.

ODS-EA-OnlyOS and ODS-EA-OnlyDS. Specifically, ODS-EA-OnlyOS is the same algorithm as ODS-EA, while the improving strategy in decision space is deleted. Similarly, the only difference between ODS-EA and ODS-EA-OnlyDS is that the evolutionary strategy in objective space is removed in ODS-EA-OnlyDS. Note that to ensure a fair comparison, all three algorithms adopt the same number of evaluations.

The comparison results are reported in terms of error rate, feature subset size, and HV values, which are shown in Table 9. From the table, we can find that compared with its two variants, ODS-EA has better performance, no matter measured by error rate, feature subset size, or HV values. The win/tie/loss results also justify the superiority of ODS-EA. This fact demonstrates the effectiveness of the two suggested strategies in the objective space and decision space, respectively. In addition, further analysis shows that since the feature contribution-based improving strategy in the decision space is removed, ODS-EA-OnlyOS cannot remove the redundant features effectively, which degrades the performance of ODS-EA-OnlyOS. Similarly, due to the subregion-based evolutionary strategy in the objective space, the search of ODS-EA-OnlyDS is still performed in a feature space with high dimensions, which leads to the size of the obtained feature subsets is large. By combining the above analysis, we can conclude that both strategies suggested in our ODS-EA are effective and complementary. Specifically, the subregion-based evolutionary strategy in the objective space solves the “curse of dimensionality” by guiding the search space toward promising regions, while the feature contribution-based improving strategy in the decision space further improves the quality of the feature subsets. Moreover, the

**Table 8**
The average HV values obtained by different MOEAs.

| Datasets | BDGMOEA | MOFS-MST | ODS-EA |
|---|---|---|---|
| ORL | **0.851**±0.026(≈) | 0.837±0.026(≈) | 0.844±0.021 |
| colon | 0.859±0.059(≈) | 0.859±0.057(≈) | **0.871**±0.063 |
| warpPIE10P | 0.946±0.037(+) | 0.954±0.039(≈) | **0.962**±0.026 |
| lymphoma | 0.769±0.090(+) | 0.792±0.085(+) | **0.813**±0.068 |
| GLIOMA | 0.800±0.071(+) | 0.819±0.072(+) | **0.861**±0.076 |
| DLBCL | 0.943±0.070(+) | 0.947±0.066(+) | **0.968**±0.057 |
| TOX_171 | 0.820±0.036(+) | 0.811±0.037(+) | **0.880**±0.034 |
| Brain1 | 0.623±0.132(+) | 0.615±0.117(+) | **0.678**±0.113 |
| ALLAML | 0.963±0.059(+) | 0.965±0.054(+) | **0.989**±0.030 |
| Brain2 | 0.786±0.081(+) | 0.785±0.097(+) | **0.828**±0.086 |
| CLL_SUB_111 | 0.816±0.058(+) | 0.781±0.071(+) | **0.844**±0.036 |
| GLI_85 | 0.885±0.060(+) | 0.859±0.078(+) | **0.913**±0.049 |
| +/ ≈ /− | 10/2/0 | 9/3/0 | – |

**Table 9**
The comparison results between ODS-EA and its two variants.

| Datasets | ODS-EA-OnlyOS | ODS-EA-OnlyDS | ODS-EA |
|---|---|---|---|
| ORL | Err: 0.196±0.036(+)<br>Size: 88.7±46.8(+)<br>HV: 0.813±0.024(+) | Err: 0.142±0.028(+)<br>Size: 111.4±30.8(+)<br>HV: 0.804±0.026(+) | **Err: 0.132**±0.023<br>**Size: 56.8**±10.7<br>**HV: 0.844**±0.021 |
| colon | Err: 0.233±0.084(+)<br>Size: 14.8±6.7(+)<br>HV: 0.834±0.049(+) | Err: 0.150±0.064(≈)<br>Size: 24.9±14.0(+)<br>HV: 0.859±0.057(≈) | **Err: 0.142**±0.067<br>**Size: 10.9**±4.9<br>**HV: 0.871**±0.063 |
| warpPIE10P | Err: 0.075±0.047(+)<br>Size: 19.0±5.4(+)<br>HV: 0.938±0.039(+) | Err: 0.117±0.040(+)<br>Size: 237.1±108.8(+)<br>HV: 0.828±0.044(+) | **Err: 0.038**±0.028<br>**Size: 16.2**±5.4<br>**HV: 0.962**±0.026 |
| lymphoma | Err: 0.211±0.084(≈)<br>Size: 97.4±39.1(≈)<br>HV: 0.800±0.076(+) | Err: 0.214±0.083(≈)<br>Size: 182.6±274.5(≈)<br>HV: 0.782±0.084(+) | **Err: 0.202**±0.076<br>**Size: 80.9**±76.3<br>**HV: 0.813**±0.068 |
| GLIOMA | Err: 0.230±0.100(+)<br>Size: 25.6±12.4(+)<br>HV: 0.804±0.074(+) | Err: 0.185±0.097(≈)<br>Size: 56.7±57.7(+)<br>HV: 0.829±0.086(+) | **Err: 0.153**±0.086<br>**Size: 17.1**±15.1<br>**HV: 0.861**±0.076 |
| DLBCL | **Err: 0.030**±0.050(≈)<br>Size: 27.2±31.4(+)<br>**HV: 0.972**±0.045(≈) | Err: 0.053±0.062(≈)<br>Size: 177.4±148.3(+)<br>HV: 0.931±0.069(+) | Err: 0.035±0.065<br>**Size: 17.0**±18.2<br>HV: 0.968±0.057 |
| TOX_171 | Err: 0.274±0.054(+)<br>Size: 108.1±37.3(+)<br>HV: 0.796±0.036(+) | Err: 0.168±0.043(+)<br>Size: 352.5±115.2(+)<br>HV: 0.807±0.044(+) | **Err: 0.124**±0.039<br>**Size: 69.0**±33.6<br>**HV: 0.880**±0.034 |
| Brain1 | Err: 0.446±0.113(+)<br>Size: 168.1±192.4(+)<br>HV: 0.604±0.106(+) | Err: 0.373±0.123(≈)<br>Size: 18.4±12.5(≈)<br>HV: 0.677±0.109(≈) | **Err: 0.363**±0.121<br>**Size: 15.7**±5.3<br>**HV: 0.678**±0.113 |
| ALLAML | Err: 0.022±0.037(≈)<br>Size: 53.9±27.3(+)<br>HV: 0.979±0.034(+) | Err: 0.090±0.071(+)<br>Size: 356.5±270.4(+)<br>HV: 0.881±0.077(+) | **Err: 0.011**±0.033<br>**Size: 18.0**±19.7<br>**HV: 0.989**±0.030 |
| Brain2 | Err: 0.309±0.138(+)<br>Size: 53.4±41.4(+)<br>HV: 0.773±0.114(+) | Err: 0.202±0.101(≈)<br>Size: 36.1±29.6(+)<br>HV: 0.822±0.082(≈) | **Err: 0.189**±0.095<br>**Size: 18.3**±8.4<br>**HV: 0.828**±0.086 |
| CLL_SUB_111 | Err: 0.324±0.090(+)<br>Size: 102.8±90.2(+)<br>HV: 0.770±0.060(+) | Err: 0.216±0.078(≈)<br>Size: 87.0±81.4(+)<br>HV: 0.814±0.066(+) | **Err: 0.200**±0.050<br>**Size: 29.7**±8.3<br>**HV: 0.844**±0.036 |
| GLI_85 | Err: 0.155±0.087(+)<br>Size: 128.3±76.0(+)<br>HV: 0.877±0.068(+) | Err: 0.140±0.061(+)<br>Size: 745.0±569.1(+)<br>HV: 0.848±0.065(+) | **Err: 0.095**±0.054<br>**Size: 46.4**±26.8<br>**HV: 0.913**±0.049 |
| +/ ≈ /− | Err: 9/3/0<br>Size: 11/1/0<br>HV: 11/1/0 | Err: 5/7/0<br>Size: 10/2/0<br>HV: 9/3/0 | –<br>–<br>– |

searches in the objective space and decision space are performed sequentially, which ensures that the goals of both stages are fully achieved. In this way, ODS-EA achieves the better trade-off among error rate, feature subset size, and HV values.

Besides the above experiments, we further investigate the performance of the proposed ODS-EA through a series of supplementary studies. Due to space limitations, we present them in the Supplementary Materials. Specifically, these experiments include: 1) The comparison between the proposed ODS-EA and baselines on the high-dimensional datasets with more instances; 2) The comparison between the proposed ODS-EA and more FS algorithms; 3) The comparison between the proposed ODS-EA and its two single-stage hybrid variants; 4) The sensitivity analysis of parameters $\theta$ and $thr$ in the proposed ODS-EA; and 5) The stability analysis between the proposed ODS-EA and baselines.

## 5. Conclusion and future work

In this paper, we tackled the HDFS problem from a new perspective: objective space. Specifically, an objective to decision space-based evolutionary algorithm, termed ODS-EA, was proposed. In ODS-EA, a subregion-based evolutionary strategy was first designed, which could not only solve the "curse of dimensionality", but also obtain latent good feature subsets effectively. Then, a feature contribution-based improving strategy was developed, which was performed in the decision space and further enhanced the quality of the final feature subsets. Experimental results on different high-dimensional datasets demonstrated the superiority of the proposed ODS-EA.

The performance of our ODS-EA shows that it is a promising way to tackle the HDFS problem by utilizing the search in the objective space. However, this work is only the first step, there still exist many interesting issues to be further investigated. For example, different from this work that combines two space search in a sequential manner, in the future, we plan to combine them in an evolutionary multitasking (EMT) framework, where objective space search and decision space search are treated as two tasks. By designing the useful knowledge transfer between these two tasks, we can solve the "curse of dimensionality" and obtain high-quality feature subsets effectively. In addition, the basic evolutionary paradigm used in our ODS-EA is GA. However, as reviewed in Section II, there are many other EC paradigms (e.g., PSO and DE) that solve HDFS effectively. How to integrate the proposed method with other EC paradigms, so that the performance of ODS-EA can be further enhanced, is also an interesting future work.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Fan Cheng reports financial support was provided by National Natural Science Foundation of China. Kaixuan Li reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.eswa.2025.130560.

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